

Preconditioned Crank–Nicolson sampling for spinDMFT

1 Setting

The spin-DMFT self-consistency equations require the thermal spin correlations

$$g^{\alpha\beta}(\tau) = \frac{\int dV p(V) \text{Tr}[U(\beta-\tau, \beta; V) S^\alpha U(\tau, 0; V) S^\beta]}{\int dV p(V) Z(V)}, \quad (1)$$

where $V = \{V^\alpha(\tau_n)\}$ is the mean-field noise trajectory living in the diagonal Matsubara frequency basis, $p(V) = \prod_{n,\alpha} \mathcal{N}(0, \lambda_{n,\alpha})$ is a Gaussian prior with eigenvalues $\lambda_{n,\alpha}$ determined self-consistently from the covariance matrix, $U(\tau_2, \tau_1; V)$ is the imaginary-time propagator generated by the single-spin Hamiltonian $H(\tau; V) = V(\tau) \cdot S$, and $Z(V) = \text{Tr}[U(\beta, 0; V)]$ is the per-trajectory partition function.

Defining the target measure

$$\pi(dV) \propto p(V) Z(V) dV, \quad (2)$$

equation (1) can be written as an expectation under π :

$$g^{\alpha\beta}(\tau) = \mathbb{E}_\pi \left[\frac{\text{Tr}[U(\beta-\tau, \beta; V) S^\alpha U(\tau, 0; V) S^\beta]}{Z(V)} \right]. \quad (3)$$

An unbiased Monte Carlo estimator therefore accumulates the per-sample observable $\text{Tr}[\dots]/Z(V)$ under draws $V^{(i)} \sim \pi$.

2 Difficulties with random-walk Metropolis–Hastings

The simplest Markov chain targeting π uses Gaussian random-walk proposals $V' = V + \varepsilon \xi$, $\xi \sim p(V)$, with Metropolis–Hastings acceptance probability

$$a(V \rightarrow V') = \min \left(1, \frac{p(V') Z(V')}{p(V) Z(V)} \right). \quad (4)$$

In the diagonal Matsubara frequency basis the prior factorises, so the proposal noise is drawn component-wise as $\xi_{n,\alpha} \sim \mathcal{N}(0, \lambda_{n,\alpha})$. The step size ε is tuned for an acceptance rate of roughly 25%.

The fundamental limitation of random-walk M–H is that its autocorrelation time scales as $\tau_{\text{int}} = \mathcal{O}(d)$, where d is the dimension of V . Here $d = 3N_\tau$ (three spin components times the number of time steps). For $N_\tau = 50$ this gives $d = 150$, so only $N/\tau_{\text{int}} \approx N/150$ effectively independent samples are obtained from N M–H steps. Because the importance-sampling estimator produces *independent* draws from $p(V)$, random-walk M–H offers no practical advantage at moderate N_τ .

3 Preconditioned Crank–Nicolson (pCN)

The pCN algorithm [1] is designed precisely for targets of the form $\pi \propto p(V) \Phi(V)$ where p is Gaussian and $\Phi(V) = Z(V) > 0$. Its proposal reads

$$\boxed{V' = \sqrt{1 - \beta^2} V + \beta \xi, \quad \xi \sim p(V),} \quad (5)$$

with $\beta \in (0, 1]$ a tunable step-size parameter. The acceptance probability simplifies to

$$a(V \rightarrow V') = \min\left(1, \frac{Z(V')}{Z(V)}\right), \quad (6)$$

because the Gaussian factors in $p(V')$ and $p(V)$ cancel exactly against the proposal densities. The key properties are:

1. **Dimension-independent mixing.** For targets of the form $\pi \propto p \cdot \Phi$ the acceptance rate and autocorrelation time of pCN are independent of d , in contrast to the $\mathcal{O}(d)$ scaling of random-walk M–H. This makes pCN the preferred algorithm in high-dimensional Gaussian-prior settings.
2. **No prior-ratio computation needed.** In the random-walk variant the term $p(V')/p(V)$ requires evaluating $\log p$, which involves summing over all d components. In pCN this ratio is absorbed into the proposal by construction, so only $Z(V')$ and $Z(V)$ need to be evaluated. Since a Z -evaluation requires a full imaginary-time propagation (the bottleneck of the simulation), pCN and random-walk M–H cost the same per step while pCN mixes faster.
3. **Single tuning parameter.** The scalar β controls the proposal step size. $\beta = 1$ recovers independence M–H (draw V' fresh from the prior), $\beta \rightarrow 0$ gives infinitesimal moves. Optimal performance is typically achieved at β corresponding to an acceptance rate of 20–40%.

4 Implementation in the diagonal Matsubara basis

Working in the basis that diagonalises the per-block covariance matrix (with eigenvalues $\lambda_{n,\alpha}$), the prior is a product of independent Gaussians. In this basis the pCN proposal (5) acts component-wise:

$$V'_{n,\alpha} = \sqrt{1 - \beta^2} V_{n,\alpha} + \beta \xi_{n,\alpha}, \quad \xi_{n,\alpha} \sim \mathcal{N}(0, \lambda_{n,\alpha}). \quad (7)$$

After proposing V' in the diagonal basis one applies the orthogonal back-rotation and the inverse Fourier transform to obtain the time-domain field $V'(\tau_n)$, which is then passed to the propagator routine. The chain state and the acceptance step (6) are otherwise identical to the random-walk variant already implemented in `Algorithm_MH/`.

The main loop changes relative to the random-walk implementation are:

1. Replace the random-walk proposal $V' = V + \varepsilon \xi$ with $V' = \sqrt{1 - \beta^2} V + \beta \xi$.
2. Remove the $\log p(V')/p(V)$ term from the log-acceptance ratio; only $\log Z(V') - \log Z(V)$ remains.
3. Rename the CLI parameter `--mhStepSize` to `--pcnBeta` (or repurpose it) with a default of, e.g., $\beta = 0.5$.

5 Expected performance gain

For the random-walk variant with $d = 150$ and $N = 80\,000$ nominal samples the estimated autocorrelation time is $\tau_{\text{int}} \approx d/(2.38^2 \times 0.234) \approx 115$ steps, giving an effective sample size $\text{ESS} \approx 700$. Under pCN with the same N and a comparable acceptance rate the autocorrelation time is $\mathcal{O}(1)$ in d , so $\text{ESS} \approx N$, representing a gain of roughly two orders of magnitude at $N_\tau = 50$. The gain grows linearly with N_τ , making pCN increasingly advantageous at finer imaginary-time discretisations.

References

- [1] S. L. Cotter, G. O. Roberts, A. M. Stuart, D. White, *MCMC methods for functions: modifying old algorithms to make them faster*, Statistical Science **28**, 424 (2013).